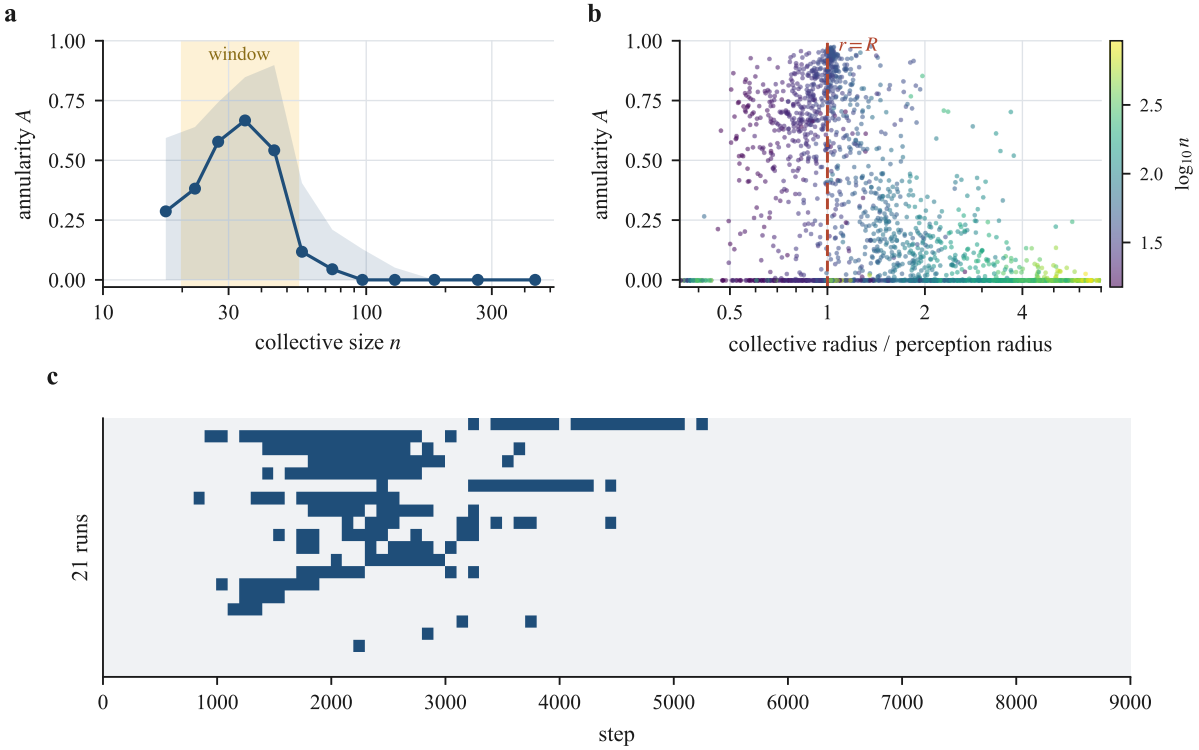

a
1.00
0.75
0.50
0.25
0.00
annularity A
window
10
30
100
300
collective size n
b
1.00
0.75
0.50
0.25
0.00
annularity A
r = R
0.5
1
2
4
collective radius / perception radius
2.5
2.0
1.5
log10 n
c
21 runs
0
1000
2000
3000
4000
5000
6000
7000
8000
9000
step